

A disproof of Conjecture 00000007664

$$9\Gamma_{1/4}(3/2) + 8\Gamma_{1/4}(-1/2) = 0$$

1 The conjecture

Conjecture 00000007664 reads:

Definition: The Jackson q -Gamma function is $\Gamma_q(x) = (1-q)^{1-x}(q; q)_\infty / (q^x; q)_\infty$, where $(a; q)_\infty$ is the infinite q -Pochhammer product. *Conjecture:* For every algebraic q in $(0, 1)$ and any two distinct positive integers m, n , the numbers $\Gamma_q(m/n)$ and $\Gamma_q(1 - m/n)$ are algebraically independent over $\mathbb{Q}(q)$, except when $m/n = 1/2$.

The Chinese text gives $(a; q)_\infty = \prod_{k \geq 0} (1 - aq^k)$.

Reading. Literally: for every algebraic real $q \in (0, 1)$ and all positive integers $m \neq n$ with $m/n \neq 1/2$, the two real numbers $\Gamma_q(m/n)$, $\Gamma_q(1 - m/n)$ are algebraically independent over the field $\mathbb{Q}(q) \subset \mathbb{R}$. That is, no nonzero $P \in \mathbb{Q}(q)[X, Y]$ vanishes at the pair. The text allows $m > n$. We give a counterexample at which both values are finite and nonzero, every infinite product converges and no denominator vanishes. The counterexample therefore also refutes the restricted reading “whenever both values are defined”.

2 Definitions and sources

Definition 1. For real a, q : $(a; q)_\infty := \lim_{N \rightarrow \infty} \prod_{k=0}^{N-1} (1 - aq^k)$, and $\Gamma_q(x) := (1 - q)^{1-x}(q; q)_\infty / (q^x; q)_\infty$ with real powers. This is the definition in the statement and in Wikipedia (https://en.wikipedia.org/wiki/Q-gamma_function): $\Gamma_q(x) = (1 - q)^{1-x} \prod_{n \geq 0} \frac{1 - q^{n+1}}{1 - q^{n+x}} = (1 - q)^{1-x} \frac{(q; q)_\infty}{(q^x; q)_\infty}$ for $|q| < 1$. The page also records the functional equation $\Gamma_q(x + 1) = \frac{1 - q^x}{1 - q} \Gamma_q(x)$. The proof below re-derives the instance it needs directly from the products, so this equation is not assumed.

3 The counterexample

Take $q = 1/4$, $m = 3$, $n = 2$. Then q is rational, so it is algebraic and $\mathbb{Q}(q) = \mathbb{Q}$. We have $m \neq n$, $m/n = 3/2 \neq 1/2$ and $1 - m/n = -1/2$. Moreover $q^{3/2} = 1/8$, $q^{-1/2} = 2$ and $1 - q = 3/4$.

Lemma 2 (Convergence). *If $0 \leq a \leq 1$ and $0 \leq q < 1$, the partial products $P_N = \prod_{k < N} (1 - aq^k)$ are non-increasing and non-negative, so they converge, and $(a; q)_\infty \geq 1 - a/(1 - q)$. Hence $A := (1/4; 1/4)_\infty \geq 2/3$ and $L := (1/8; 1/4)_\infty \geq 5/6$.*

Proof. Each factor lies in $[0, 1]$, so (P_N) is non-increasing and non-negative. By induction, $P_N \geq 1 - a \sum_{k < N} q^k$. Indeed $P_{N+1} = P_N(1 - aq^N) \geq (1 - aS_N)(1 - aq^N) \geq 1 - aS_N - aq^N$, where $S_N = \sum_{k < N} q^k \leq 1/(1 - q)$. $\square$

Lemma 3 (Shift). $(2; 1/4)_\infty$ exists and equals $-\frac{1}{2}L$.

Proof. For every N : $\prod_{k < N+2} (1 - 2q^k) = (1 - 2)(1 - \frac{1}{2}) \prod_{k < N} (1 - 2q^{k+2}) = -\frac{1}{2} \prod_{k < N} (1 - \frac{1}{8}q^k)$, since $2q^{k+2} = q^k/8$. Let $N \rightarrow \infty$ and use Lemma 2. $\square$

Theorem 4. With $q = 1/4$: $\Gamma_q(3/2) = (3/4)^{-1/2}A/L > 0$, $\Gamma_q(-1/2) = (3/4)^{3/2}A/(-\frac{1}{2}L) < 0$, and

$$9\Gamma_q(3/2) + 8\Gamma_q(-1/2) = 0.$$

Hence the nonzero polynomial $P(X, Y) = 9X + 8Y \in \mathbb{Q}(q)[X, Y]$ vanishes at $(\Gamma_q(3/2), \Gamma_q(-1/2))$. These two numbers are not algebraically independent over $\mathbb{Q}(q)$, and Conjecture 00000007664 is false.

Proof. The formulas for the two values follow from the definition, $q^{3/2} = 1/8$, $q^{-1/2} = 2$ and Lemma 3. Their signs follow from $A, L > 0$. Then $9\Gamma_q(3/2) + 8\Gamma_q(-1/2) = \frac{A}{L}(9(3/4)^{-1/2} - 16(3/4)^{3/2}) = 0$, because $(3/4)^{3/2} = (3/4)^2(3/4)^{-1/2} = \frac{9}{16}(3/4)^{-1/2}$. P is nonzero because $P(1, 0) = 9$. $\square$

Remark 5 (Scope). (a) The relation is the functional equation applied twice: $\Gamma_q(3/2) = [1/2]_q[-1/2]_q \Gamma_q(-1/2)$, with $[x]_q = (1 - q^x)/(1 - q)$. The same argument works whenever m/n is a half-integer $\geq 3/2$, because then m/n and $1 - m/n$ differ by an integer. For $m/n \in \mathbb{Z}$, $m/n \geq 2$, the number $1 - m/n$ is a pole. (b) The phrase “except when $m/n = 1/2$ ” suggests that the author may have meant $m < n$, so that $m/n \in (0, 1)$. Under that reading the statement is a genuine transcendence question, and it is *not* addressed here. The text says “any two distinct positive integers”, and we refute that literal statement.

4 Lean 4 formalization

The directory `lean/` contains a Lean 4 project (Lean v4.33.1, Mathlib v4.33.1). `Conjecture7664.lean` imports the single source file `Conjecture7664/Basic.lean` (236 lines, namespace `C7664`).

Definitions.

- `qPochPartial a q N := prod k in range N, (1 - a * q ^ k)`.
- `qPochInf a q := limUnder atTop (qPochPartial a q)` (the limit of the partial products).
- `qGamma q x := (1 - q) ^ (1 - x) * qPochInf q q / qPochInf (q ^ x) q` with `Real.rpow`.
- **Conjecture:** for all `q : Real` with `IsAlgebraic Q q`, $0 < q < 1$, and all `m n : Nat` with $0 < m$, $0 < n$, $m \neq n$, $(m:\mathbb{R})/n \neq 1/2$: `AlgebraicIndependent (IntermediateField.adjoin Q {q}) ![qGamma q (m/n), qGamma q (1 - m/n)]`. This uses Mathlib’s `AlgebraicIndependent` over the subfield $\mathbb{Q}(q)$ of $\mathbb{R}$.

Theorems. `tendsto_partial` (Lemma 2), `shift_tendsto`, `shift_eq` (Lemma 3), `gamma_three_halves`, `gamma_neg_half`, `linear_relation` : $9 * \text{qGamma } (1/4) (3/2) + 8 * \text{qGamma } (1/4) (-1/2) = 0$, `counterexample_values_genuine`: the products $(q; q)_\infty$, $(q^{3/2}; q)_\infty$ and $(q^{-1/2}; q)_\infty$ at $q = 1/4$ converge (`Tendsto`), with $A > 0$, $(q^{3/2}; q)_\infty > 0$ and $(q^{-1/2}; q)_\infty < 0$, and $\Gamma_q(3/2) > 0 > \Gamma_q(-1/2)$. This shows that no junk value of `limUnder` or of division by zero is involved. `not_algebraicIndependent` is Theorem 4 with $P = 9X_0 + 8X_1$. The main theorem is

`conjecture_00000007664_false` : $\neg$ Conjecture

It is proved by instantiating $q = 1/4$, $m = 3$, $n = 2$.

Axiom audit. There is no `sorry`, no `native_decide` and no added axiom. The file compiles with no warnings.

'C7664.conjecture_00000007664_false' depends on axioms:

[`propext`, `Classical.choice`, `Quot.sound`]

'C7664.counterexample_values_genuine' depends on axioms:

[`propext`, `Classical.choice`, `Quot.sound`]

To check: `cd lean; lake exe cache get; lake build`.